

Snake Bite – Envenomation

Guideline Title: Snake Bite - Envenomation

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- I. **PURPOSE:** To recognize the patient with a snake bite and envenomation, provide adequate supportive care, appropriate interventions (antivenom therapy if necessary, from the referring facility), and emergent treatment and expeditious transport to the nearest appropriate medical facility.
- II. **PATHOPHYSIOLOGY:** Snake venom varies depending on the species of snake. When possible, identification of the snake is important to provide the most appropriate treatments.
 - a. Elapidae: Coral snakes, sea snakes, cobra, brown snakes. “Red on yellow, kill a fellow”.
 - i. Neurotoxic, no swelling, 10-14h onset, nausea, vomiting, sweating, lethargy
difficulty speaking, decreased respiratory, cardiac arrest.
 - b. Crotalidae (Pit Viper): rattle snakes, cottonmouths/ water moccasins, copper heads.
 - i. Histamine release, rapid swelling, local tissue destruction.
- III. **DEFINITIONS:**
- IV. **DEPARTMENT GUIDELINE:**
 - a. Assess and manage airway, breathing, and circulation.
 - b. Treat dysrhythmias per *GUID-C001-C010 Cardiac Guidelines*.
 - c. If signs of inadequate perfusion treat per *GUID-M011 - Hypotension*.
 - d. Treat allergic reaction per *GUID-M004 Allergic Reaction Anaphylaxis* or *GUID-PED002 Pediatric Allergic Reaction Anaphylaxis*.
 - e. Treat pain and/or anxiety per *GUID-G005 – Analgesia and Sedation Management*.
 - f. Remove restrictive clothing/jewelry and control bleeding. Immobilize extremity of possible. Note fang or bite marks. Note with pen on the skin the level of inflammation / induration. Make note of serial circumferential measurements of extremity every 15 minutes.
 - g. Do not handle or transport the snake, dead or alive. Take a picture of the snake for identification. Poison center callers can now send photos of their bite and/or the snake so that our poison specialists can immediately determine the type of snake involved (1-800-222-1222).
 - h. Keep area of injury below level of the heart. Do not apply ice.
 - i. Distal pulses should be checked and monitored.

- j. Transport patient to facility with anti-venom capabilities. If bite is visible to upper or lower extremities with high suspicion of venomous snake, transport to a state approved trauma center.
- k. For inter-facility transport, when anti-venom is being administered, follow manufacturer guidelines for administration.
 - i. **Crotalidae Immune FAB Antivenins (Anavip) or Antivenin (Crotalidae):**
 1. 1 vial dissolves in 10ml of NS. Dilute total amount to a 1:2 or 1:4 concentration. 5-10 vials if moderate bite. 5-20 vials for a severe bite. Start at 1ml/min for 10 minutes to assess for reaction then increase rate to give over 1 hour. For pediatric patients' rate is 10ml/kg/hour.
 2. Assess for allergy to papaya, papain, pineapples, sheep, horses, dust mites, or latex.
 - i. **Crofab:** for crotalid envenomation.
 1. Given within 6 hours. 4-12 vials in 250ml based on clinical judgement. Start at 50ml/h for the first 10 minutes to assess for an allergic reaction, then increase to 250ml/h. Then administer 2 vials every 6 hours for next 18 hours. Pediatric dose = adult dose.
 2. Reconstitute each vile with 18ml of 0.9% NaCl slowly into the vial. Rotate vial 180 degrees and manually invert up to twice per second until no solid material remains in the vial. No not shake. The entire dose should then be further diluted in NA to a final total volume of 250ml for infusion.
 3. Do not administer CroFab® if known hypersensitivity to papaya or papain or to any of its components unless the benefits outweigh the risks.
 - l. Do not obtain IV lines, blood pressuring monitoring, or labs from the bitten extremity.
 - m. Contact the receiving physician, poison control, or medical director prior to the initiation of antivenom.

V. **DOCUMENTATION:** EMS Charts

VI. **REFERENCES:**

- a. <https://crofab.com/>
- b. Ashton, J., Baker, S., & Weant, K. (2011). When Snakes Bite: The Management of North American Crotalinae Snake Envenomation. *Advanced Emergency Nursing Journal*, (33)1, 15-22.
doi: 10.1097/TME.0b013e318207cc27

VII. **APPENDIX:**

- a. **Nonvenomous snakes** **Venomous snakes**



- b. **NONVENOMOUS SNAKES:** Head is usually oval, but may be somewhat triangular, pupils round, no pits-only nostrils present, divided scales on underside of tail. Although many snakes vibrate their tail when upset, nonvenomous snakes never have rattles
- c. **VENOMOUS SNAKES:** Head distinctly triangular, pupils elliptical, pits as well as nostrils present, undivided scales on underside of tail.